

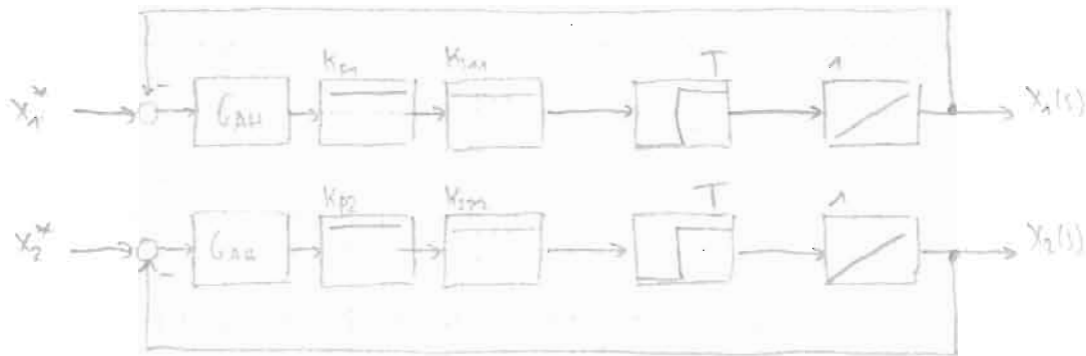
Aufg. 4

(1. Streckenteil / verkoppelter Teil)

(a)

$$\left. \begin{array}{l} G_{s11} = K_{s11} \\ G_{s21} = K_{s21} K_{s22} \end{array} \right\} \Rightarrow \underline{G}_s = \begin{bmatrix} K_{s11} & K_{s12} \\ K_{s22} K_{s21} & K_{s22} \end{bmatrix}$$

$$\underline{G}_K = \frac{1}{\det \underline{G}_s} \begin{bmatrix} G_{s11} G_{s22} & -G_{s12} G_{s22} \\ -G_{s21} G_{s21} & G_{s11} G_{s22} \end{bmatrix} = \frac{1}{K_{s11} K_{s22} - K_{s22} K_{s21} K_{s12}} \begin{bmatrix} K_{s11} K_{s22} & -K_{s12} K_{s22} \\ -K_{s21} K_{s21} K_{s22} & K_{s11} K_{s22} \end{bmatrix}$$



(b) beide Kreise relativ identisch, Entwurf gemeinsam, nachher Trennung

Regler:

$$G_R(z) = K_P$$

Strecke: (inkl. GA1)

$$G_S(s) = \frac{1-e^{-sT}}{s} \cdot K_S e^{-sT} \cdot \frac{1}{s} \Rightarrow G_S(z) = \frac{z-1}{z} z^{-1} \cdot \mathcal{Z}\left\{\frac{K_S}{s^2}\right\}$$

$$G_O(z) = G_R(z) \cdot G_S(z) = \frac{K_P K_S T}{z(z-1)} = \frac{z_0}{N_0}$$

$$= \frac{z-1}{z} z^{-1} K_S \frac{Tz}{(z-1)^2} = \frac{K_S T}{z(z-1)}$$

charak. Polynom:

$$N(z) = z_0 + N_0 = \underbrace{K_P K_S T}_{c} - \underbrace{z}_{b} + \underbrace{z^2}_{a} \stackrel{!}{=} 0$$

$$z_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

aperiodisch, wenn " $\sqrt{=0}$ " -

$$\Rightarrow b^2 \stackrel{!}{=} 4ac \Leftrightarrow 1 \stackrel{!}{=} 4 K_P K_S T \Rightarrow K_P = \frac{1}{4 K_S T}$$

Kreis 1:

$$K_{P1} = \frac{1}{4 K_{s11} T}$$

Kreis 2:

$$K_{P2} = \frac{1}{4 K_{s22} T}$$